CARBYNESENTINEL
Implementation Plan

Final Year Project — BS Computer Science
National University of Computer and Emerging Sciences, Lahore
Adeena Riaz (23L-2501) · Ali Asad Khan (23L-0802) · Maryam Amanat Ali Butt (23L-0779)
Supervisor: Usman Aamer

Version 2.0 — November 2026 status correction. *Internal working document. Black and white. Synced to master roadmap Steps 1–15 and the dataset repository evidence. Public paper stays conservative; this plan is the full internal build target delivered in* ***FYP-1 (to end of November 2026)*** *then* ***FYP-2****.*

1. Purpose of this plan

This document tells the three-person team what to build, in what order, why each piece exists in research terms, what already exists in the dataset repo, and how we will know a stage is done. It does not assign named owners.

CarbyneSentinel is a **passive** pre-confirmation watcher on an Ethereum test network (e.g. Sepolia). Pending mempool traffic is admitted, EVM bytecode is turned into a CFG and opcode sequence, a GCN and a LoRA-tuned Qwen2.5-Coder score structure and semantics, a manipulation fingerprint flags deliberately disguised code, and a fused risk score raises a human review flag. We do not cancel transactions, halt validators, or mutate chain state.

Latency rule (corrected): the hard **500 ms pre-confirmation SLA** applies to the fast non-LLM path (Bloom + Fingerprint + GCN). The LLM stage is **excluded** from that hard guarantee; on timeout, fall back to the fast-path verdict / Fail-Closed policy.

2. System picture

2.1 Online path (runtime) — Contract Adapter — design target

1. Listen to pending transactions on Sepolia (WebSocket / RPC) — **FYP-2**.

2. Counting Bloom filter: drop duplicates / absorb spam — **FYP-2**.

3. Bytecode reader: disassemble to opcodes; build CFG — **FYP-1 Nov prototype**.

4. Manipulation fingerprint: score deliberate disguise — **FYP-1 Nov Fingerprint v0**.

5. GCN structural detector on CFG — **FYP-2** (full train).

6. LoRA Qwen2.5-Coder semantic / intent auditor — **FYP-2**.

7. Fusion of structural + semantic + fingerprint → risk in [0, 1] — **FYP-2**.

8. If score ≥ threshold: flag + log; async Firebase alert. Else: log only — **FYP-2 wiring**.

UI prototype for alerts/dashboard already exists in `frontend/` with seeded data (Step 10 partially done).

2.2 Parallel adapters

- Contract Adapter (Step 7): core FYP path — FYP-2.
- AgentGuard (Step 8): optional FYP-2; do not claim FYP-1.
- Bridge Adapter (Step 9): design-only this cycle.
- Dashboard (Step 10): React prototype done; live scores FYP-2.

2.3 Offline path (training and datasets)

Processed corpora exist. Obfuscation benchmark scaled. Unique-seed Phases 1–4 locked. DeFi-Intent 150-seed done. Train full GCN + LoRA + fusion + full evaluation = **FYP-2**. FYP-1 November adds feature prototype + **baseline** smoke train only.

2.4 Out of scope

Mainnet funds, private key custody, automated on-chain pause/cancel, full Bridge product, claiming Forta vendor SLAs as our results.

3. Research framing: gaps we fill

No single reviewed system combines counting Bloom filter intake, bytecode CFG + GCN, grounded LLM checking, and a timed pre-confirmation path. Paper-to-action map (unchanged intent):

Paper / source	What it gives us	Our action
Haq et al. Carbyne (2025)	CBF mempool admission idea	Build CBF intake (FYP-2); admission only
SmartBugs / SolidiFI	Shared gold / injected labels	Primary detection benches + aug seeds
Zhuang / SCVHunter GNNs	GCN for contract vulns	GCN on bytecode CFG (FYP-2); not Solidity IR
GPTScan / ORACAL / FinDet	LLM + grounding ideas	Short opcode prompts + GCN second vote
OBFPROBE (2025)	Obfuscation breaks detectors	Verified obfuscation bench (done) + fingerprint
Flash Boys / DeFiRanger	Pending stage + intent	DeFi-Intent-Bench seed (done); enrich FYP-2
Forta docs	Industry pre-inclusion existence	Cite as motivation; never paste vendor %

4. Model choices (locked for v1 — training mostly FYP-2)

Structural: 2–3 layer GCN over bytecode CFG (PyG); class-weighted loss.
Semantic: Qwen2.5-Coder + LoRA (smallest size that fits hardware); 4/8-bit OK; stub fallback for CI.
Fingerprint + Fusion: third signal; late-fuse 25% / 32% / 43% (design); ablations required.
FYP-1 only requires a **baseline** feature model for smoke train — not the full GCN/LoRA stack.

5. Data status (what is already done) — corrected

Corpus	Role	Approx. size	Status
SolidiFI + SmartBugs master	Gold / injected; compile path	493 vuln	Done (466/493 compiled)
master_dataset_phase2.json	Extended master after Phase 2	+304 parents	Done (promote as default Nov)
DIVE vulnerable / benign	Scale / rare-class mining	~19.6k / ~2.7k	Done (JSONL pools)
DAppSCAN	Audit-based fill	~5k	Done
DeFiHackLabs	Real exploit PoCs	~848	Done
Slither HF benign	Benign pool	~35.1k	Done
DeFi-Intent-Bench	Exploit vs transfer vs arb	150 seed (50/50/50)	Seed done; enrich FYP-2
Obfuscation benchmark	Disguised + re-verified pairs	401 verified / 142 failed / 192 stacked	Done at scale
Phases 1–4 packaging	Unique-seed balance + splits	720 @ 80; 504/108/108	Done; leakage-checked
Phase 5 smoke train	Baseline / full models	—	Baseline = FYP-1 Nov; full GCN/LLM = FYP-2

5.1 Class imbalance (corrected)

Raw master was skewed (e.g. unchecked_call 152 vs denial_of_service 6). After Phase 2 grow + Phase 3 hard-cap, **v1 train is balanced: 9 classes × 80 = 720 (ratio 1.0)**. No further balancing is required for counts. Optional FYP-2: improve label quality on DIVE-mined rares / add more true benigns.

5.2 Schema for originals and transforms

All records follow `target_schema.json`. Critical transform fields: `parent_contract_id`, `transformation_level`, `transformation_techniques`, `obfuscation_severity`, `exploit_verified_after_transform`.

6. Phased implementation plan — rebaselined November 2026

Old v1.1 weeks assumed GCN/LoRA/Sepolia inside ~12 weeks from plan authoring. ***Rebaseline:*** *treat data Phases as largely complete; finish a thin FYP-1 feature/smoke slice by end of November; move heavy ML + live path to FYP-2.*

Phase A — DONE (data factory; do not redo)

Schema freeze; process SolidiFI/SmartBugs + pools; obfuscation ladder + verify/fail logs; Phases 1–4 unique-seed packaging; DeFi-Intent 150 seed; class distribution reports. Acceptance: packaging artifacts committed; Phase 4 leakage check green.

Phase B — FYP-1 November remaining (must finish by end of November 2026)

Goals

1. Promote phase2 master as default.
2. Disassembler: hex → (pc, mnemonic, immediates).
3. Basic blocks + CFG edges; export JSON (+ optional PyG Data).
4. Fingerprint v0: Shannon entropy + jump-ratio features.
5. Baseline smoke train on 504/108/108 using fingerprint/opcode features (sklearn or tiny MLP) → log one macro-F1.
6. Keep React UI demoable on seeded data.

Acceptance (FYP-1 bar)

• Disasm+CFG run without crash on a documented small compiled batch (aim ≥90% of that batch).
• Fingerprint features written for the same batch.
• Smoke train completes and writes baseline macro-F1 JSON/log.
• No claim of trained GCN/QLoRA/fusion/live Sepolia.

Phase C — FYP-2 (after November)

Sub-phase	Build	Acceptance sketch
C1 Bloom intake	CBF by tx/bytecode hash; flood test	Dups rejected; not a latency bottleneck
C2 Harden features	CFG over full compiled train; cache graphs/opcodes	Crash rate <10% on compiled master
C3 Train GCN	PyG loader; weighted CE; early stop on val macro-F1	Beats majority baseline; metrics committed
C4 LoRA Qwen	PEFT LoRA; short opcode prompts; stub fallback	Adapter loads; val metrics + LLM latency logged separately
C5 Fusion	25/32/43 or learned fuse; threshold sweep; ablations	Ablation table; fused not worse than best single
C6 Sepolia + UI wire	WS listen + reconnect; replay fixtures; Firebase async	Replay e2e alerts; live soak optional
C7 Full eval	Detection / ablation / evasion / intent / latency tracks	Results chapter numbers from logs only

Sub-phase	Build	Acceptance sketch
C8 Demo + publish	Demo polish; optional Zenodo/HF; AgentGuard if time	Defense-ready demo <3 min setup

7. Module checklist (engineering) — status column added

Module	Inputs → Outputs	FYP status
mempool_listener	WS RPC → raw txs	FYP-2
cbf_intake	tx/code hash → admit/reject	FYP-2
disasm_cfg	bytecode → CFG + opcodes	FYP-1 Nov prototype
fingerprint	opcodes/stats → disguise score	FYP-1 Nov v0
gcn_model	PyG graph → score	FYP-2
llm_lora	opcode text → score/label	FYP-2
fusion	gcn+llm+fp → risk	FYP-2 (UI mock exists)
augment_obf	seed → variant + schema	DONE (401/142)
phase1–4 packaging	master/pools → 720 + splits	DONE
baseline_smoke	features → macro-F1 log	FYP-1 Nov
eval_runner	splits → metrics JSON	FYP-2 (full); Nov baseline only
alerter / dashboard	risk → log + Firebase / UI	UI prototype DONE; live scores FYP-2
agentguard	agent payment intent → risk	Optional FYP-2

8. Suggested repository layout

Create when Phase B/C code lands (not all folders exist yet):
`carbynesentinel/{intake,features,models/gcn,models/llm,fusion,augment,eval,configs,scripts}`
Keep `dataset/` raw immutable; write only under processed/, reports/, checkpoints.

9. Latency budget (working split) — SLA corrected

Stage	Budget (ms)	Notes
Intake + CBF	10–20	Must stay tiny; part of 500 ms fast path
Disasm + CFG + opcodes	40–80	Cache when re-seeing code
Fingerprint	~12–32	Part of 500 ms fast path
GCN infer	20–85	Part of 500 ms fast path
LLM LoRA infer	variable	**Outside hard 500 ms SLA**; grey-band / timeout fallback
Fusion + log	5–15	Always cheap

If LLM alone blows any soft budget: early-exit; quantise; smaller Qwen; run LLM only when GCN score is in a grey band; never block pre-confirmation triage waiting on LLM.

10. Training recipe (practical)

Labels: binary vulnerable/benign; primary_vulnerability_type; intent labels only on DeFi-Intent-Bench. Avoid leakage: parent+transforms same split; no threshold tune on test. Baselines to beat: majority class; opcode n-gram/RF; later GCN-only / LLM-only. **FYP-1:** ship the classical/tiny baseline. **FYP-2:** GCN/LLM/fuse.

11. How augmentation builds our dataset

Quality rule unchanged: verified transforms preferred; always set parent_contract_id. Ladder lexical → structural → control_flow → data_flow → contract_level (+ stacks) is implemented. First-release acceptance (≥200 verified) is already exceeded (401).

12. Risks and mitigations

Risk	Mitigation
LLM latency > budget	Exclude from hard 500 ms SLA; grey-band; quantise; timeout fallback
Class imbalance	Already capped for v1; weighted loss if reopened
CFG incompleteness	Mark unresolved edges; batch failure CSV; fallback features
Scope creep (Forta clone)	Watcher only; FYP-1 thin slice; FYP-2 sequence
Overclaiming “done” detectors	Use §16 status table; no invented metrics
Sepolia RPC flaky	Replay fixtures first; reconnect backoff

13. Milestone calendar (rebaselined)

13.1 Through end of November 2026 — FYP-1

Window	Focus	Exit check
Already complete	Data factory Phases 1–4 + obfuscation + intent seed + UI prototype	Artifacts in repo; 720@1.0; 401/142; React demo
Remaining Nov days	Promote phase2 master; disasm+CFG; Fingerprint v0; baseline smoke F1	Batch CFG success logged; baseline macro-F1 file written
FYP-1 report	Ch.7 prototype narrative; Ch.8 packaging (+ optional baseline F1); Ch.9 plan	No fake GCN/LLM accuracy tables

13.2 After November 2026 — FYP-2 (indicative)

Block	Focus	Exit check
1	Bloom + harden features over full train cache	CBF flood test; CFG crash <10%
2	GCN train + polish	Val macro-F1 > majority; checkpoint frozen
3	LoRA Qwen train/eval	Adapter + separate LLM latency log
4	Fusion + ablations	Ablation table committed
5	Replay e2e (+ optional Sepolia) + UI live scores	Alert JSON; latency histogram
6	Full eval campaign + demo + report sync	Results from logs only; defense demo

14. Working agreements (team)

Do not edit raw/ datasets. Do not invent evaluation numbers. Schema fields mandatory on augmented rows. Prefer small verified batches. If blocked >48 h, log and switch tasks. Advisor updates: short bullets + one figure max.

15. Definition of project done

FYP-1 done when: data factory + UI prototype + Nov feature prototype + baseline smoke F1 + honest report milestones.

Full FYP done when: runnable replay/Sepolia Contract Adapter with fused flags; GCN+LoRA+fingerprint+fusion+ablations; detection metrics on SmartBugs/SolidiFI; latency measured with SLA distinction; obfuscation evasion track on verified pairs; intent eval on DeFi-Intent-Bench; dashboard with real/replay scores; report+defense. AgentGuard and dataset publish optional.

16. Master roadmap sync — Steps 1–15 corrected status

Step	Name	Track	Corrected status	Plan phase
1	Bytecode reader	A	FYP-1 Nov prototype	Phase B
2	Manipulation fingerprint	A	FYP-1 Nov Fingerprint v0	Phase B
3	Structural detector	A	CFG Nov; full GCN FYP-2	B then C3
4	Semantic LLM (+ intent)	A	Intent seed done; QLoRA FYP-2	C4
5	Fusion (3 signals)	A	UI mock; calibrate FYP-2	C5
6	Bloom intake	B	FYP-2	C1
7	Contract Adapter	B	FYP-2	C6
8	AgentGuard	B	Optional FYP-2	C8
9	Bridge adapter	B	Design-only	Design note
10	Dashboard	B	React prototype DONE; live scores FYP-2	Done shell / C6
11	Obfuscation benchmark	A	DONE 401/142/192	Phase A
12	DeFi-Intent-Bench	A	150 seed DONE	Phase A
13	Evaluation	A	Baseline F1 Nov; full campaign FYP-2	B / C7
14	Dataset publishing	A	Not started	C8 optional
15	Report & defense	A+B	FYP-1 in progress	Ongoing

17. Requirements traceability (short)

FR bytecode→CFG (FR-2.x) → Phase B. FR GCN/LLM/fusion (FR-3.x) → Phase C. FR mempool + CBF + 500 ms (FR-1 / NFR) → Phase C with SLA excluding LLM. Dashboard UI shell exists; live alert wiring → Phase C6.

18. Minimum test matrix (keep; timeline corrected)

Test ID	Pass rule	When
T-CFG-01/02	Opcode list non-empty; edges or unresolved marked	FYP-1 Nov
T-CFG-03	Crash rate <10% on compiled master	FYP-2 harden
T-FP-01	Entropy/jump features for batch	FYP-1 Nov
T-BASE-01	Baseline smoke train writes macro-F1	FYP-1 Nov
T-CBF-01/02	Second hash rejected; flood bounded	FYP-2
T-GCN-01/02	Overfit sanity; val > majority	FYP-2
T-LLM-01/02	Loss drops; LLM p95 recorded separately	FYP-2
T-FUS-01	Ablation table produced	FYP-2
T-AUG-01	Verified transform flag set (already at scale)	Done
T-E2E-01/02	Replay alerts valid; optional live soak	FYP-2

19. Decision log (v1 locks + Nov corrections)

Decision	Choice
LLM family	Qwen2.5-Coder + LoRA (FYP-2 train)
Graph model	Simple GCN on bytecode CFG (FYP-2 train)
Hot-path IR	Bytecode only
Primary benches	SmartBugs + SolidiFI
Augmentation role	Rebalance (done for v1 counts) + evasion set (401+)
Output policy	Flag/log only (passive watcher)
500 ms SLA	Fast non-LLM path only; LLM excluded
FYP-1 Nov bar	Disasm+CFG+Fingerprint v0+baseline smoke F1 — not full neural stack
Bloom / Sepolia / AgentGuard	FYP-2 (AgentGuard optional)

20. Anti-patterns (do not do these)

Training on test hashes. Reporting Forta vendor percentages. Calling CBF rejects “attacks blocked.” Claiming 500 ms without a measured histogram. Claiming 500 ms includes LLM. Augmenting without parent_contract_id. Treating unverified transforms as evasion proof. Writing “GCN/QLoRA/Bloom already built” without repo evidence. Inventing Ch.8 accuracy tables.

21. End-to-end narrative (vision — not all live yet)

A user submits a contract-related transaction on Sepolia. While it sits pending, our listener receives the hash and fetches the payload. The counting Bloom filter asks whether we already spent model time on this hash; if yes, we stop. If no, we disassemble bytecode into opcodes and a CFG. Fingerprint scores disguise look. The GCN reads the graph. LoRA-tuned Qwen reads a truncated opcode prompt. Fusion turns signals into one score. If high, we raise a review flag. We never cancel the transaction. Offline, we improve models using SmartBugs/SolidiFI and verified obfuscation pairs so detectors do not collapse the way OBFPROBE showed.

That story is the product. FYP-1 makes the data + UI + first features/smoke true. FYP-2 makes the full online hybrid path true with logs and numbers.

22. Printable acceptance checklist

[x] Data env / processed corpora / schema
[x] Obfuscation ≥200 verified (actually 401) + fail log
[x] Train/val/test manifests (Phase 4) + leakage check
[x] Class distribution after cap = 1.0
[x] React dashboard prototype (seeded)
[ ] Disasm+CFG batch prototype (FYP-1 Nov)
[ ] Fingerprint v0 batch (FYP-1 Nov)
[ ] Baseline smoke macro-F1 log (FYP-1 Nov)
[ ] CBF duplicate test (FYP-2)
[ ] GCN beats majority baseline (FYP-2)
[ ] LoRA adapter + val metrics (FYP-2)
[ ] Ablation table (FYP-2)
[ ] Replay e2e alert JSON + latency histogram (FYP-2)
[ ] Detection + evasion metrics from logs (FYP-2)
[ ] Demo script < 3 minutes setup (FYP-2)

23. What to say if asked “what is novel?”

1. Pending-path hybrid screener (CBF + bytecode CFG/GCN + LoRA code LLM + fingerprint) under an explicit latency budget — proposed stack; trained fusion is FYP-2.
2. Bytecode-first graphs matching mempool reality.
3. Original fingerprint idea: flag deliberate disguise.
4. Verified obfuscation benchmark (done) + DeFi-Intent-Bench seed (done).
5. Passive watcher ethics: flag for humans, no automated chain harm.

24. Closing note to the team

We already have a large, schema-checked data foundation, a scaled obfuscation benchmark, balanced seed-family splits, an intent seed, and a React operator console prototype. By end of November, finish the thin feature/smoke slice. After November, build Bloom, trained GCN, LoRA Qwen, three-signal fusion, replay/Sepolia adapter, full evaluation, and defense polish. Keep research gaps visible so the final defence can point from paper → gap → module → metric. Bring logs, not slides.

*End of implementation plan. Document version: **2.0** — November 2026 status correction (synced to master roadmap Steps 1–15 + FYP-1/FYP-2 split + 500 ms SLA excluding LLM). Black and white by design.*